AMRITA
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Department of Computer Science and Engineering

23CSE498 **Project Phase II** **2026**

B.Tech CSE — Initial Project Idea Submission

Project Team

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1. Project Title

Project Title:

Explainable AI-Based Element Recommendation Framework for Functional Materials Using Unsupervised Representation Learning

2. Project Abstract

2.1 Motivation

Motivation:

The search for replacement materials for things like batteries, semiconductors and catalysts is a tough process. It usually relies on knowledge, lots of trial and error and computer simulations that cost a lot. Recently Artificial Intelligence and Materials Informatics have helped us use machine learning to understand relationships and guess material properties. Many current methods do not give clear reasons for their suggestions and do not validate them properly.

This project wants to create an AI tool that helps make decisions about materials. It will use a type of machine learning that finds patterns on its own, suggest materials in a way that makes sense and check these suggestions against existing materials databases and scientific research. The goal is to give materials researchers clear scientifically sound advice. The tool will use information that's already available to the public. It aims to make materials research more efficient and reliable.

2.2 Problem Statement

Develop an AI-based explainable element recommendation system that identifies chemically similar elements using unsupervised machine learning and recommends potential substitute elements for functional materials. The system should provide reasoning for each recommendation, validate generated substitute compounds using publicly available materials databases and published literature, and indicate when further computational or experimental validation is required.

3. Objectives

Objectives:

1. Collect and integrate elemental properties and functional materials data from publicly available databases to build the project dataset.
2. Develop an unsupervised machine learning model to identify chemically similar elements.
3. Design an AI-based recommendation system for suggesting substitute elements in functional materials.
4. Generate explainable recommendations and validate them using materials databases and published literature.

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5. Develop a decision-support interface that presents recommendations, explanations, validation evidence, and confidence levels.

4. Dataset / Data Source (if applicable)

Dataset / Data Source:

The new system uses information that's available to the public about materials science to help learn about similar elements make good recommendations and validate things based on evidence. It gets information about elements from places like Mendeleev, XenonPy and pymatgen. It also gets information about compounds, material properties and structures from the Materials Project and the Open Quantum Materials Database. These datasets are good, which the system is reliable and easy to understand and works the same every time, which makes it good, for evaluating things.

1. **Mendeleev (Element Properties)**
   Website: https://mendeleev.readthedocs.io/
2. **XenonPy (Material Descriptors)**
   GitHub: https://github.com/yoshida-lab/XenonPy
3. **Pymatgen (Python Materials Genomics)**
   GitHub: https://github.com/materialsproject/pymatgen
4. **Open Quantum Materials Database (OQMD)**
   Website: https://oqmd.org/

5. Expected Deliverables

Expected Deliverables:

1. An Explainable AI-based Element Recommendation System for identifying chemically similar elements and recommending substitute elements for functional materials.
2. A Validation Module that explains recommendations and verifies them using materials databases and published literature with confidence scores.
3. A Research Report and Software Prototype demonstrating the implementation, evaluation, and results of the proposed framework.

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6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	**PNcsp+: A Periodic Number-Based Crystal Structure Prediction Method Enhanced by Machine Learning**	2026	Introduces PNcsp+, which uses the Mendeleev Periodic Number as a single transparent similarity descriptor to generate candidate crystal structures via elemental substitution, then ranks them using an ensemble of pretrained GNN energy predictors (MACE, M3GNet, ALIGNN-FF). Achieves state-of-the-art accuracy (86.1% space-group match, 85% structure match in Top-5) on the CSPBench benchmark without needing structure relaxation, and shows the approach generalizes to hybrid organic–inorganic perovskites.
2	**Recreation of the periodic table with an unsupervised machine learning algorithm**	2021	Develops the Periodic Table Generator (PTG), an unsupervised algorithm based on generative topographic mapping (GTM-LDLV) that autonomously reconstructs Mendeleev-like tabular arrangements of elements purely from 39 physicochemical features, with no prior chemical labels. Demonstrates that the resulting element coordinates serve as an effective, low-dimensional descriptor for downstream formation-energy prediction, outperforming PCA/t-SNE-based descriptors.
3	**Atom2Vec: learning atoms for materials discovery**	2018	Proposes Atom2Vec, which learns dense atom embeddings purely from the co-occurrence statistics of atoms and their compositional "environments" across a materials database (analogous to word embeddings in NLP), using SVD (model-free) or likelihood optimization (model-based). Shows these unsupervised vectors recover known periodic-table groupings and outperform empirical descriptors in formation-energy prediction for elpasolite and half-Heusler compounds.

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No.	Paper Title	Year	Key Contribution
4	**Learning Hidden Chemistry with Deep Neural Networks**	2021	Presents DeepOFM, a deep-learning model that scores the likelihood of an atom occupying a given local chemical environment (Voronoi-based), and uses this to build an atom-substitution recommendation system. Introduces a dissimilarity measure (Jensen–Shannon divergence over recommendation distributions) that recovers known chemical groupings (e.g., lanthanides vs. transition metals), and validates 71 novel substituted structures in $Nd_2Fe_{14}B$ via DFT.
5	**Compositionally restricted attention-based network for materials property predictions (CrabNet)**	2021	Introduces CrabNet, which applies Transformer self-attention to structure-agnostic materials property prediction directly from chemical composition. Matches or exceeds prior best-in-class models (Roost, ElemNet) across 28 benchmark property-prediction datasets, and shows that attention maps and per-element contribution scores offer a step toward model interpretability without requiring crystal structure input.
6	**Phase Prediction via Crystal Structure Similarity in the Periodic Number Representation**	2024	Introduces the original PNcsp algorithm, which predicts stable crystal prototypes for chemical systems by searching up to third-nearest neighbors in the Mendeleev Periodic Number (PN) phase map and screening candidates through DFT-based thermodynamic, mechanical, and dynamical stability checks. Applied to 59 unreported equiatomic binary systems, the method identified 93 candidate prototypes, 47 of which are fully stable, including 19 entirely novel polymorphic phases. The approach was further validated on 15 binary, ternary, and quaternary benchmark phases (CSPBench) using CrystalNN structural similarity scoring, correctly reproducing most experimental target structures.

Table 1: Preliminary Literature Survey

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6.1 Comparison of Related Work

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Oran et al., "PNcsp+: A Periodic Number-Based Crystal Structure Prediction Method Enhanced by Machine Learning," J. Chem. Theory Comput.	2026	Template-based crystal structure prediction using Mendeleev Periodic Number (PN) as a single interpretable similarity descriptor; candidate structures scored/ranked using ensembled GNN energy predictors (MACE, M3GNet, ALIGNN-FF)	OQMD (1.4M+ structures), Materials Project, MPDS/Linus Pauling File, CSPBench (180 systems)	Substitution is driven by one hand-picked numeric descriptor (PN distance) rather than a learned multi-property similarity space; "reasoning" for a recommendation is just a scalar distance, not a human-readable explanation; validation of stability still needs GNN relaxation/DFT; no mechanism to say "no evidence exists yet" for a compound outside the template library
2	Kusaba et al., "Recreation of the periodic table with an unsupervised machine learning algorithm," Scientific Reports	2021	Unsupervised GTM-LDLV (generative topographic mapping) maps 39 physicochemical features of elements onto 2D/3D grids; learned table used as an element descriptor for downstream formation-energy regression	XenonPy features for 54 elements (H–Xe), Materials Project formation energies	Purely a representation-learning/visualization exercise – it clusters elements but never proposes or explains a substitution; no compound-level validation pipeline; capped at 54 elements; no natural-language justification, only heatmap overlays

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No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
3	Zhou et al., "Atom2Vec: learning atoms for materials discovery," PNAS	2018	Unsupervised atom-environment co-occurrence matrix + SVD (model-free) or likelihood optimization (model-based) to learn dense atom vectors; vectors clustered/evaluated on downstream property prediction	Materials Project inorganic compounds (binary/ternary/quate ~60k formulas)	Similarity is captured only composition-ranllayry, (structure is explicitly ignored); no explicit "recommend substitute X for Y" output – only vector proximity; no reasoning text; no cross-check against literature or database for whether a substituted compound is real or previously reported
4	Nguyen et al., "Learning Hidden Chemistry with Deep Neural Networks," arXiv	2021	Supervised/self-supervised Deep-OFM: MLP encoders for center atom + Voronoi-weighted environment, trained to score local-structure likelihood; used to recommend substitute atoms and generate new candidate structures	OQMD (lanthanide/transit metal/light-element compounds, ~4,200 structures → 24,494 local environments); case study on $Nd_2Fe_{14}B$	Recommendation ioisn- a bare likelihood/probability score with no chemical-language justification; validating substitutions still requires full DFT relaxation (expensive); no automated check of whether the generated candidate already exists in a public database before recommending; dissimilarity metric (Jensen–Shannon) is post-hoc, not surfaced to the end user as a reason

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No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
5	Wang et al., "Compositionally restricted attention-based network for materials property predictions (CrabNet)," npj Comput. Mater.	2021	Transformer self-attention over element-derived matrix (EDM) built from element embeddings (mat2vec) + fractional stoichiometry encodings; per-element contribution and attention-map visualization for interpretability	28 benchmark datasets (Matbench + Extended: MP, AFLOW, OQMD, experimental band gaps, etc.)	Designed for property prediction, not substitution recommendation; interpretability is limited to attention-weight visualization, which is not equivalent to a human-readable chemical justification; no mechanism for proposing alternative elements or validating them against databases
6	Oran et al., "Phase Prediction via Crystal Structure Similarity in the Periodic Number Representation," *Inorg. Chem.*	2024	Uses the Mendeleev Periodic Number (PN) to identify chemically similar elements and generate candidate crystal structures through elemental substitution. Candidate structures are validated using DFT-based thermodynamic, mechanical and dynamical stability analysis.	Linus Pauling File (LPF), Materials Project, OQMD, CSP-Bench	Uses PN as a fixed similarity descriptor rather than learned representations; lacks human-readable explanations; relies on computationally expensive DFT validation; no automated evidence-based validation using public databases or scientific literature.

Table 2: Comparison of related work on interpretable materials substitution and structure prediction.

Comparison of related work on interpretable materials substitution and structure prediction.

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6.2 Novelty / Research Gap Statement

Based on what we have read current studies have done a job in learning about elements, predicting material properties and suggesting new materials. However they only focus on one part of the problem. Some methods, like PTG and Atom2Vec are good at understanding chemical similarities. They do not explain why they suggest certain materials or check if they really work. Some other methods, like PNcsp+ and DeepOFM try to find materials or substitute atoms but they rely on complex computer checks and do not tell apart suggestions that are proven to work and those that are just ideas. There are also methods, like mat2vec that learn from papers but they only look at what words appear together not the actual properties of materials. There is a gap in current research. We need a system that can learn about material similarities suggest materials, with clear reasons and check if they work using available data and scientific papers. The proposed work aims to fill this gap by combining these parts into a decision-support system. This system will provide evidence-based suggestions and identify cases that need more checks. It will help make material discovery more transparent and efficient. The system will use available materials databases and scientific literature to support its suggestions. By doing it will make it easier to find new materials that really work.

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7. Potential Publication Venues

Conferences

1. **IEEE International Conference on Data Mining (ICDM) Workshops**
   Suitable for AI, machine learning, recommendation systems, and data-driven materials applications.

2. **IEEE International Conference on Artificial Intelligence (IEEE CAI)**
   Covers Explainable AI, intelligent systems, and AI applications in scientific domains.

3. **IEEE International Conference on Big Data**
   Focuses on data analytics, machine learning, and scientific data mining.

4. **ACM Conference on Recommender Systems (RecSys) – Workshop Track**
   Relevant because the project develops an explainable recommendation framework.

5. **International Conference on Materials Informatics (MIn) / AI for Materials Workshops**
   Specifically focuses on AI and machine learning applications in materials science.

Journals

1. **npj Computational Materials (Nature Portfolio)**
   One of the leading journals for AI-driven materials discovery and materials informatics.

2. **Computational Materials Science (Elsevier)**
   Publishes work on computational methods, machine learning, and materials applications.

3. **Journal of Materials Informatics**
   Dedicated to data-driven materials research and AI-based materials discovery.

4. **Materials Today Communications**
   Accepts interdisciplinary work involving AI and computational materials research.

5. **Scientific Reports (Nature Portfolio)**
   Suitable for interdisciplinary AI and materials science research.

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8. References

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[5] A. Y.-T. Wang, S. K. Kauwe, R. J. Murdock, and T. D. Sparks, "Compositionally Restricted Attention-Based Network for Materials Property Predictions," *npj Computational Materials*, vol. 7, p. 77, 2021. Available: https://doi.org/10.1038/s41524-021-00545-1

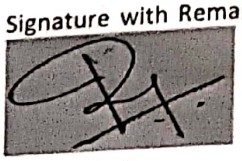
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9. Faculty Remarks

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